[i]APPLIED PROCESS ENGINEERING

GROUP-6

NAME:

P. MOUNI SREELATHA(CH24B025)

M. SATYA SREE(CH24B024)

K. SRI ANUSHKA(CH24B041)

CH. KAVYA SRI(CH24B009)

भारतीय प्रौद्योगिकी संस्थान तिरुपति

TIRUPATI

INDIAN INSTITUTE OF TECHNOLOGY TIRUPATI

DEPARTMENT OF CHEMICAL ENGINEERING

INDEX

1. ABSTRACT

The objective of this study is to design a continuous flow process capable of producing 50 tons/day of anhydrous ammonia using the Haber–Bosch process. Nitrogen is supplied from air, and hydrogen is generated via steam reforming of natural gas. The analysis applies material and energy balance principles to determine input and output requirements, recycle-purge specifications, and the operating conditions for each unit are specified in the flow diagram. Industrial practice shows that ammonia formation is favoured at high pressures (200 bar) and moderate temperatures (400-500°C), while an Fe-based catalyst is employed for enhancement of the reaction. A complete flow diagram was designed, including reactors, coolers, preheaters, and compressors. The material balances were carried out in kmol/day units while energy balances consider only for temperature changes. The final design achieves the target production rate of 50 tons of ammonia per day, requiring approximately 1430.8 kmol/day of natural gas for hydrogen generation and 2138.6 kmol/day of air for nitrogen supply. A recycle ratio of 4.5 and purge ratio of 2.25% were selected to limit accumulation of inert gases to maximise conversion in the synthesis loop. Secondary streams include CO_2 from reforming, vapour and unreacted gases (argon, CH_4) that appear in the purge stream. These by-products can be recovered and utilized as fuel gas or for water recovery. This design demonstrates a feasible approach for industrial–scale ammonia production.

2. INTRODUCTION

Ammonia is one of the most widely produced industrial chemicals worldwide. It is a key component in fertilizers, chemicals, and industrial processes. It is very useful on regular basis; despite having large usage, production of ammonia is complicated because of its raw materials. Large-scale production of ammonia requires high efficiency, energy management, and safe operation. Optimizing operating conditions are crucial for the production [1].

There are different types of process for ammonia preparation. Industrially via the Haber–Bosch process reacting nitrogen with hydrogen at high pressure and temperature with an Iron catalyst. Other methods include the hydrolysis of metal nitrides and emerging green routes like electrochemical synthesis from nitrogen and water, aiming for more sustainable production [2].

The production of ammonia is generally dominated by Haber–Bosch process because it is highly optimized, economically feasible and kinetics using high pressure and moderate temperature(400 – 650°C), and iron catalyst, allowing large scale and continuous production. But the alternative methods struggle with economic feasibility and separation process. Carl Bosch scaled Fritz Haber's lab discovery, creating robust equipment for high pressures(150-400 atm) and integrating continuous recycling for maximum efficiency [3].

The Haber–Bosch process is the primary method for industrial ammonia synthesis. It involves reaction of nitrogen from air and hydrogen derived from natural gas under high pressure (200 bar), moderate temperature (400-500°C) with an Fe-based catalyst. Key process factors include recycle and purge ratios. This study focuses on the production of 50 tons/day of ammonia [4].

Objective of the report is to present a flow diagram representing complete ammonia production, also to analyse material, energy balances and operational conditions for maximum output.

The flow diagram starts with feed preparation from natural gas and air via desulfurizer, primary and secondary reformer, shift conversion. Then it is followed by gas purification and compression of N_2 and H_2 gases along with a flow control to maintain 1:3 ratio. It is followed by a cooler to obtain liquid ammonia.

3. ASSUMPTIONS

The following assumptions were made as per course instructions and for simplification of calculations:

- The process operates under steady-state and steady-flow conditions.
- All gaseous streams are assumed to behave as ideal gases throughout the process.
- Pressure drops in pipelines and heat exchangers are neglected.
- Energy balances consider only sensible heat changes due to temperature variation; reaction heat and shaft work are neglected as per project scope.

4. METHOD

4.a. PROCESS FLOW DIAGRAM EXPLANATION:

The overall ammonia production process is divided into several unit operations. The PFD includes all major equipment and the main material streams. Each unit is explained with operational conditions.

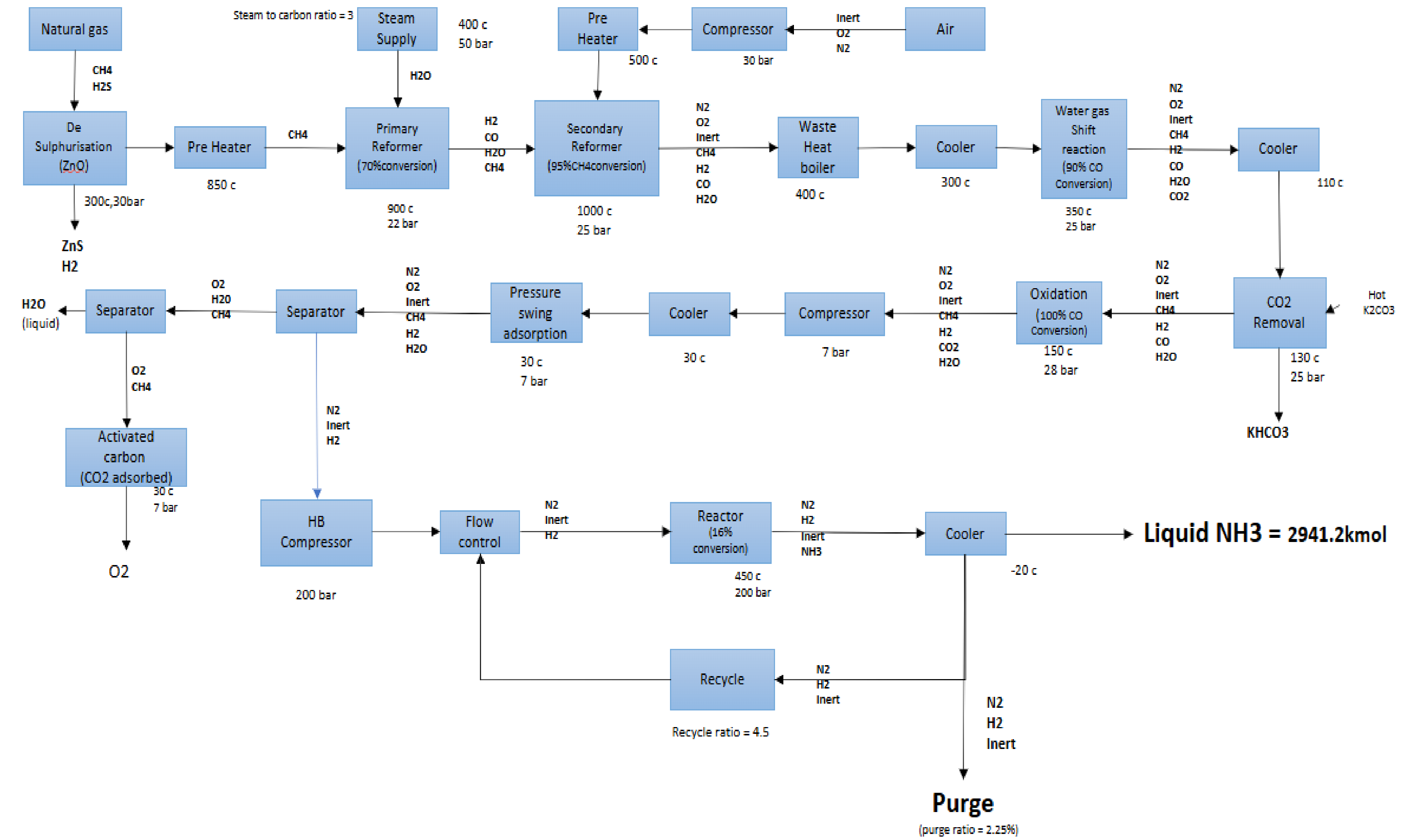

FIGURE 1: PROCESS FLOW DIAGRAM OF AMMONIA SYNTHESIS

4.b. MAJOR UNITS IN PFD:

4.b.1. DESULFURIZER: Removes sulphur and its compounds from natural gas using zinc oxide beds. Because they act as catalyst poison in further reactions, so it is necessary to remove them. It is operated at 300°C, 30 bar [5].

$$ZnO + H_2S \rightarrow ZnS + H_2O \quad \triangle Hr|_{298.25K} = -78 \text{ kJ/mol};$$

- A preheater is installed in between desulpharizer unit and primary reformer unit to raise the temperature of natural gas from 300°C to 850°C.

4.b.2. PRIMARY REFORMER: Sulphur free natural gas reacts with steam to produce CO and H_2 operates at 900°C, 22 bar. 70% single-pass conversion of natural gas is considered. A steam supply at 400°C, 50 bar and natural gas at 850°C are inputs for this unit [6].

$$CH_4 + H_2O \rightarrow CO + 3H_2 \quad \triangle Hr|_{298.15K} = 206 \text{ kJ/mol}$$

- A preheater is installed to raise the temperature of air from room temperature (25°C) to 500°C.

4.b.3. SECONDARY REFORMER: The output of primary reformer (unreacted CH_4, H_2O, CO, H_2) are reacted with O_2 from air (N_2, O_2, Ar) and produces CO. Operates at 1000°C temperature and 25 bar pressure. A single-pass conversion of 95% of natural gas is considered. The inputs from primary reformer are at 900°C, the input from air are at 500°C, whereas the outputs are at 1000°C [7].

$$CH_4 + \frac{1}{2}O_2 \rightarrow CO + 2H_2 \quad \triangle Hr|_{298.15k} = -36 \text{ kJ/mol}$$

- A cooler is added to decrease the temperature from 1000°C to 350°C and passed as an input to water gas shift reaction.

4.b.4. WATER GAS SHIFT REACTION: The output of cooler becomes the input of shift reactor which operates at 350°C, 25 bar. A single-pass conversion of 95% is considered at this operating conditions [8].

$$CO + H_2O \rightarrow CO_2 + H_2 \quad \triangle H_r|_{298.15K} = -41 \text{ kJ/mol}$$

- A cooler is added to further decrease the temperature from 350°C to 110°C to proceed with CO_2 removal

4.b.5. CO_2 REMOVAL: The output of cooler is passed on to the surface of K_2CO_3 for chemical adsorption to remove CO_2. Assumed that surface area is large enough and stream stays enough time so that all of CO_2 is absorbed by K_2CO_3 and converted into $KHCO_3$. Operates at 130°C [9].

$$K_2CO_3 + CO_2 + H_2O \rightarrow 2KHCO_3 \quad \triangle H_r|_{298.15K} = -70 \text{ kJ/mol}$$

4.b.6. OXIDATION: CO in the output of CO_2 removal gets oxidised to CO_2 by using the oxygen remaining from secondary reformer. Operates at 150°C temperature and 28 bar pressure. 100% oxidation of CO is assumed because of excess availability of O_2 [10].

$$CO + \tfrac{1}{2} O_2 \rightarrow CO_2 \quad \triangle H_r|_{298.25K} = -283 \text{ kJ/mol}$$

- A cooler is installed to reduce the temperature from 150°C to 30°C so that CO_2 adsorbs by pressure swing adsorption process

4.b.7. PRESSURE SWING ADSORPTION: Physical adsorption of CO_2 takes place at 30°C temperature and 7 bar pressure. Removing of CO_2 is mandatory because it acts as poison if it passes into Haber–Bosch process reactor [11].

- Then after a separator is placed to separate N_2, H_2, inert gases from the remaining mixture. The remaining mixture is further separated using various techniques and adsorption process to get O_2, H_2O and CH_4 and they act as energy fuels and raw products.

- The mixture of N_2, H_2, inert gases are sent into a compressor to compress from 7 bar to 200 bar. It is followed by a flow control to maintain a mole ratio of 1:3 for N_2, H_2 respectively and recycle stream mixes with the pure feed at this position [12].

4.b.8. REACTOR: The stream flow of N_2, H_2 and NH_3 flows into a reactor which operates at 450°C temperature and 200 bar pressure. A single-pass conversion of 16% is industrially approved and the output is flown into a cooler [13].

$$N_2 + 3H_2 \rightarrow 2NH_3 \quad \triangle H_{r|298.15K} = -92 \text{ kJ/mol}$$

- A cooler decreases the temperature of stream from 450°C to -20°C. Phase transfer of NH_3 occurs in this range so that we separate liquid NH_3 from real gases.
- A part of the stream of remaining gases is recycled, and the remaining is sent out as purge to prevent the accumulation of inert gases. Recycle ratio of 4.5 is used with a purge percentage of 2.25%.
- Both material and energy balances are calculated in the excel sheet attached.
- Material balance is all calculated in kmole/day units whereas energy balance is done in kJ/s units

4.c. Cp values Table

The following table is used for energy balances assuming all gases are ideal throughout the process [14]. $\sum \dot{n} Cp \triangle T = Q$

The above equation is applied to all heat exchange units considering only sensible heat changes.

$C_p{}^{ig} / R = A + BT + C\,T^2 + D\,T^{-2}$ for T(K), R=8. 314J.K^{-1}. mol^{-1}, C_p units ($Jmol^{-1}K^{-1}$)

Chemical species	A	10^3*B	10^6*C	10^{-5}*D
CH_4	1.702	9.081	-2.164	0
H_2S	3.931	1.490	0	-0.232
ZnO	45.91	9.089	4.159	-6.049
ZnS	43.11	12.658	-1.122	2.48
H_2O	3.47	1.45	0	0.121
CO	3.376	0.557	0	-0.031
H_2	3.249	0.422	0	0.083
O_2	3.639	0.506	0	-0.227
N_2	3.28	0.593	0	0.04
inert	20.786	0.000123	-0.136	0
CO_2	5.457	1.045	0	-1.157
NH_3	29.75	0.00254	-1.226	7.55

TABLE 1: C_P VALUES TABLE

- Latent heat of condensation of H_2O: -40.7 kJ/mol.
- Latent heat of condensation of NH_3: -23.5 kJ/mol.

5. RESULTS AND DISCUSSION

The following flow charts represent the complete material balance and energy balance (accounted only for temperature changes).

- 2941.2 kmol of NH_3 per day is produced by following this flow design.
- To produce 50 tons of NH_3 per day we need 1430.8 kmol of natural gas, 2138.6 kmol of air and 4163.6 kmol of H_2O is required per day.
- There is a total energy change of 11584 kJ/s to produce 50 tons of NH_3 per day.
- The overall material and energy balance for the ammonia production provides detailed information about stream compositions, temperatures, pressures, and flow rates across each unit.
- The reactor feed stream contains a nitrogen to hydrogen ratio close to 1:3, while the reactor outlet stream contains both unreacted reactants and product ammonia.
- The liquid ammonia is obtained after cooling the product stream to -20℃ temperature.
- It is an exothermic reaction. Due to thermodynamic limitations, the single pass of nitrogen remains relatively low (16%).
- This low conversion justifies the presence of a large recycle loop (recycle ratio 4.5), which improves the overall plant efficiency by returning unreacted N_2 and H_2 to the reactor.
- The purge stream ensures that inert gases do not accumulate in the system.

INPUT(KMOL/DAY)		OUTPUT(KMOL/DAY)	
NH_3	0	NH_3	2941.18
N_2	1668.08	N_2	214.23
O_2	449.10	H_2	642.66
H_2S	42.92	O_2	217.2
CH_4	1387.85	CH_4	20.8
H_2O	4163.55	H_2O	594.7
K_2CO_3	1298.68	K_2CO_3	0
ZnO	42.92	$KHCO_3$	2597.36
Inert	19.58	Inert	19.58

TABLE 2: MATERIAL BALANCE

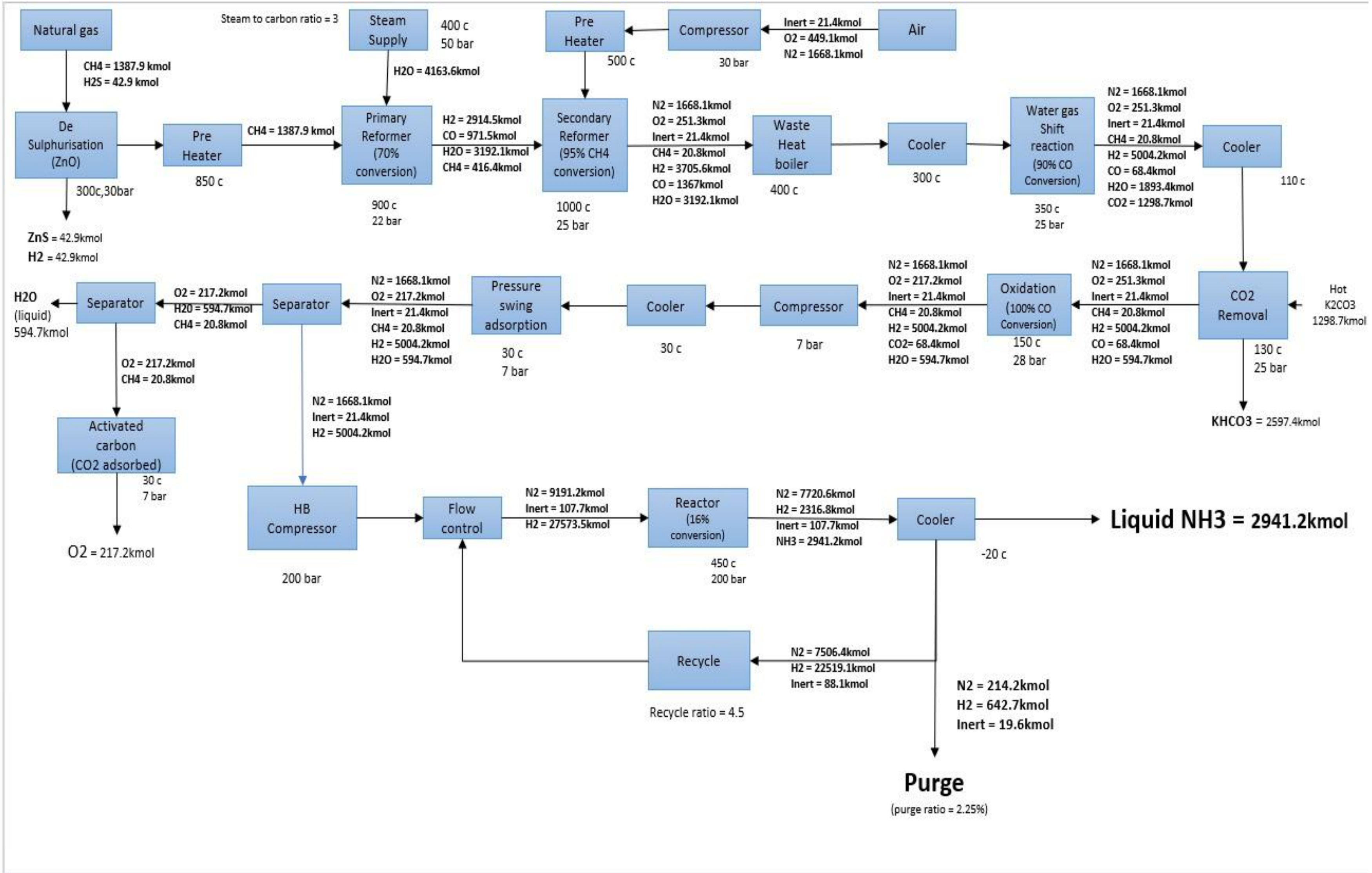

FIGURE 2- OVERALL MATERIAL BALANCE

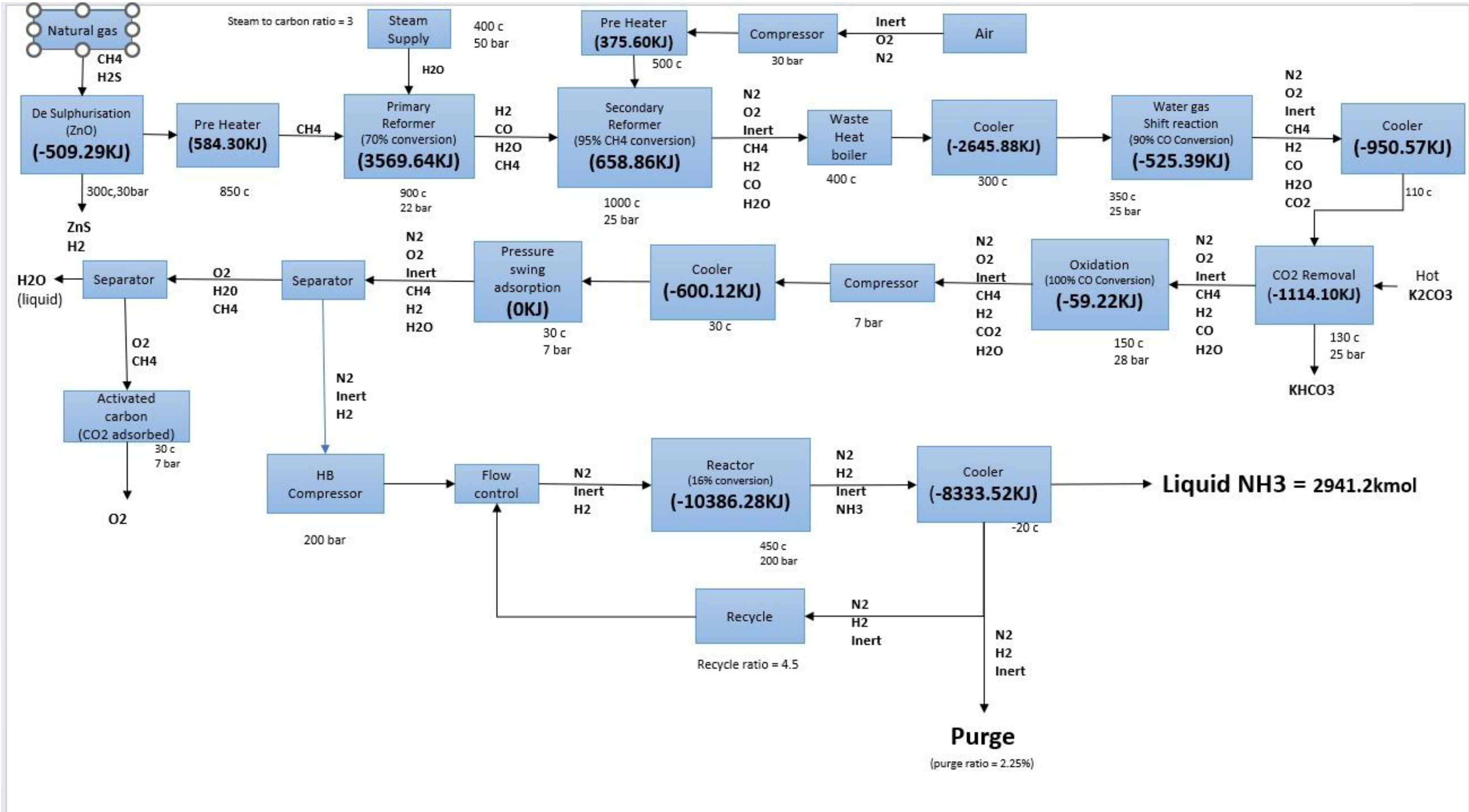

FIGURE 3- OVERALL ENERGY BALANCE(ONLY FOR TEMPERATURE CHANGE)

Sensitive analysis was performed to understand how the results vary with changes in operating conditions. When the reactor temperature was at 450°C, it is observed that lower temperature increases equilibrium by Le Chatlier's principle, but it decreases the rate of reaction. Changes in recycle ratio from 4.5 to 4 decreased the output significantly for same amount of feed, whereas increasing it from 4.5 to 5 increased the load on compressor.

6. CONCLUSION:

The purpose of this study is to design a model for synthesis of 50 tons of ammonia per day. It includes both material and energy balance (considered only for change in temperature). Assumed overall system is in steady flow and the temperatures and pressures are constant. The study shows that to produce 50 tons of ammonia, 1430.8 kmol of natural gas (95% purity is required), 4163.3 kmol of steam supply is required and 2138.6 kmol of air is required. 16% single pass conversion of Haber Bosch reactor is concluded for maximum formation of ammonia with a recycle ratio of 4.5. Although reaction mechanisms, fluid properties are not considered this study gives an overview of how material, energy balances are made and the assumptions made. A flow sheet is prepared enclosing all the work which is done.

7. CALUCULATIONS:

All material and energy balance calculations are done in the following excel sheet.

APE.xlsx

7.a. MATERIAL BALANCE:

NATURAL GAS		1430.773		DESULPHARISER					PRIMARY REFORMER			0.7		AIR	2138.564
				INPUT		OUTPUT			INPUT		OUTPUT				
	CH4	1387.85		CH4	1387.85	H2O	42.9232		H2O	4163.55	CH4	416.35		N2	1668.08
	H2S	42.9232		H2S	42.9232	ZnS	42.9232		CH4	1387.85	CO	971.5		O2	449.0985
				ZnO	42.9232	CH4	1387.85				H2	2914.5		INERT	19.58
											H2O	3192.05			
					e	42.9232				e	971.5				

	SECONDARY REFORM		0.95			WATER GAS SHIFT REACTION			0.95		CO2 REMOVAL		
INPUT		OUTPUT			INPUT		OUTPUT			INLET		OUTLET	
N2	1668.08	N2	1668.08		N2	1668.08	N2	1668.08		N2	1668.08	N2	1668.08
O2	449.0985	O2	251.3322		O2	251.3322	O2	251.3322		O2	251.3322	O2	251.3322
INERT	19.58	INERT	19.58		INERT	19.58	INERT	19.58		INERT	19.58	INERT	19.58
CH4	416.35	CH4	20.8175		CH4	20.8175	CH4	20.8175		CH4	20.8175	CH4	20.8175
CO	971.5	H2	3705.565		H2	3705.565	H2	5004.246		H2	5004.246	H2	5004.246
H2	2914.5	CO	1367.033		CO	1367.033	CO	68.35163		CO	68.35163	CO	68.35163
H2O	3192.05	H2O	3192.05		H2O	3192.05	H2O	1893.369		H2O	1893.369	H2O	594.6883
							CO2	1298.681		CO2	1298.681	KHCO3	2597.362
										K2CO3	1298.681	CO2	0
	e	395.5325				e	1298.681				e	1298.681	

OXIDATION						PREESURE SWING ADSORPTION				HB COMPRESSOR			
INLET		OUTLET			INLET		OUTLET			INLET		OUTLET	
N2	1668.08	N2	1668.08		N2	1668.08	N2	1668.08		N2	1668.08	N2	9191.176
O2	251.3322	O2	217.1564		O2	217.1564	O2	217.1564		H2	5004.245875	H2	27573.53
INERT	19.58	INERT	19.58		INERT	19.58	INERT	19.58		INERT	19.58	INERT	107.69
CH4	20.8175	CH4	20.8175		CH4	20.8175	CH4	20.8175					
H2	5004.246	H2	5004.246		H2	5004.246	H2	5004.246					
CO	68.35163	CO2	68.35163		CO2	68.35163	H2O	594.6883					
H2O	594.6883	H2O	594.6883		H2O	594.6883	CO2	0					
CO2	0	CO	0										
	e	34.17581											

REACTOR					RECYCLE	4.5		PURGE			PRODUCT
INLET		OUTLET									
N2	9191.176	N2	7720.588		N2	7506.36		N2	214.2282		NH3
H2	27573.53	H2	23161.76		H2	22519.11		H2	642.6583		2941.176
INERT	107.69	INERT	107.69		INERT	88.11		INERT	19.58		
		NH3	2941.176								
									876.4665		
						30025.47					
						670.76		inert percentage	0.02234		
	e	1470.588				0.97766					

7.b. ENERGY BALANCE:

DESULPHARISATION														
573.15	300													
		1	T	T^2	T^-2		R	8.314						
delHr		A	B	C	D									
-78			1.00E-03	0.000001	100000									
		1.702	9.081	-2.164	0		-468.05	-1.09E+03	1.17E+02	0		-1439.29	-192.215	
		3.931	1.49	0	-0.232		-1081.025	-1.79E+02	0.00E+00	-37.3351214		-1296.87	-5.35654	
		45.91	9.089	4.159	-6.04976		-12625.25	-1.09E+03	-2.24E+02	-973.571225		-14912	-61.592	-259.163816
		3.47	1.45	0	0.121		-954.25	-1.74E+02	0.00E+00	19.472197		-1108.49	-4.57848	
		43.11	12.658	-1.122	2.48		-11855.25	-1.52E+03	6.05E+01	399.099574		-12912.1	-53.3318	
		1.702	9.081	-2.164	0		-468.05	-1.09E+03	1.17E+02	0		-1439.29	-192.215	-250.1255608
													DH	**-509.2893768**
			PRIMARY REFORMER											
	del Hr	206												
673.15														
		3.47	1.45	0	0.121		-1301.25	-2.64E+02	0.00E+00	22.6084076		-1542.71	-618.081	
1123.15														
		1.702	9.081	-2.164	0		-1404.15	-5.32E+03	1.00E+03	0		-5725.34	-764.61	-1382.690861
1173.15														
		1.702	9.081	-2.164	0		1489.25	5.85E+03	-1.15E+03	0		6189.098	247.9603	
		3.376	0.557	0	-0.031		2954	3.59E+02	0.00E+00	7.75499261		3320.292	310.3955	
		3.249	0.422	0	0.083		2842.875	2.72E+02	0.00E+00	-20.7633673		3093.75	867.6521	
		3.47	1.45	0	0.121		3036.25	9.33E+02	0.00E+00	-30.2694873		3939.336	1210.01	2636.018106
													DH	**3569.635115**

			SECONDRYREFORMER									
	delHr	-36										
1173.15												
		1.702	9.081	-2.164	0	-1489.25	-5.85E+03	1.15E+03	0	-6189.1	-247.96	
		3.376	0.557	0	-0.031	-2954	-3.59E+02	0.00E+00	-7.75499261	-3320.29	-310.395	
		3.249	0.422	0	0.083	-2842.875	-2.72E+02	0.00E+00	20.7633673	-3093.75	-867.652	
		3.47	1.45	0	0.121	-3036.25	-9.33E+02	0.00E+00	30.2694873	-3939.34	-1210.01	-2636.018106
773.15												
		3.639	0.506	0	-0.227	-1728.525	-1.29E+02	0.00E+00	-46.7757643	-1904.04	-82.2839	
		3.28	0.593	0	0.04	-1558	-1.51E+02	0.00E+00	8.24242544	-1700.64	-272.976	-373.8255055
		20.786	0.000123	-0.136	0	-9873.35	-3.13E-02	1.97E+01	0	-9853.63	-18.5654	
1273.15												
		3.28	0.593	0	0.04	3198	4.54E+02	0.00E+00	-10.2742521	3641.969	584.5873	
		3.639	0.506	0	-0.227	3548.025	3.88E+02	0.00E+00	58.3063808	3993.932	96.59287	
		1.702	9.081	-2.164	0	1659.45	6.96E+03	-1.47E+03	0	7146.105	14.31509	
		3.249	0.422	0	0.083	3167.775	3.23E+02	0.00E+00	-21.3190732	3469.712	1237.212	
		3.376	0.557	0	-0.031	3291.6	4.27E+02	0.00E+00	7.9625454	3726.229	490.1677	
		3.47	1.45	0	0.121	3383.25	1.11E+03	0.00E+00	-31.0796127	4462.883	1370.823	3831.70865
		20.786	0.000123	-0.136	0	20266.35	9.42E-02	-9.24E+01	0	20174.09	38.01045	
											DH	**657.0598299**
			WATER GAS SHIFT									
	delHr	-41										
623.15												
		3.28	0.593	0	0.04	-1066	-8.88E+01	0.00E+00	6.9970655	-1147.78	-184.235	
		3.639	0.506	0	-0.227	-1182.675	-7.58E+01	0.00E+00	-39.7083467	-1298.14	-31.3953	
		1.702	9.081	-2.164	0	-553.15	-1.36E+03	1.55E+02	0	-1757.25	-3.52012	
		3.249	0.422	0	0.083	-1055.925	-6.32E+01	0.00E+00	14.5189109	-1104.58	-393.867	
		3.376	0.557	0	-0.031	-1097.2	-8.34E+01	0.00E+00	-5.42272576	-1186.01	-156.014	
		3.47	1.45	0	0.121	-1127.75	-2.17E+02	0.00E+00	21.1661231	-1323.67	-406.578	-1188.319702
		20.786	0.000123	-0.136	0	-6755.45	-1.84E-02	9.77E+00	0	-6745.7	-12.7097	
623.15												
		3.28	0.593	0	0.04	1066	8.88E+01	0.00E+00	-6.9970655	1147.782	184.2351	
		3.639	0.506	0	-0.227	1182.675	7.58E+01	0.00E+00	39.7083467	1298.137	31.39533	
		1.702	9.081	-2.164	0	553.15	1.36E+03	-1.55E+02	0	1757.248	3.520124	
		3.249	0.422	0	0.083	1055.925	6.32E+01	0.00E+00	-14.5189109	1104.584	531.9046	
		3.376	0.557	0	-0.031	1097.2	8.34E+01	0.00E+00	5.42272576	1186.012	7.800711	
		3.47	1.45	0	0.121	1127.75	2.17E+02	0.00E+00	-21.1661231	1323.665	241.1625	
		5.457	1.045	0	-1.157	1773.525	1.56E+02	0.00E+00	202.39012	2132.363	266.4771	1279.205225
		20.786	0.000123	-0.136	0	6755.45	1.84E-02	-9.77E+00	0	6745.7	12.70972	
											DH	**-525.386651**

			CO2 REMOVAL									
	del Hr	-70										
383.15												
		3.28	0.593	0	0.04	-278.8	-1.72E+01	0.00E+00	2.97629019	-292.994	-47.0297	
		3.639	0.506	0	-0.227	-309.315	-1.47E+01	0.00E+00	-16.8904469	-340.857	-8.24359	
		1.702	9.081	-2.164	0	-144.67	-2.63E+02	2.15E+01	0	-386.157	-0.77355	
		3.249	0.422	0	0.083	-276.165	-1.22E+01	0.00E+00	6.17580215	-282.208	-135.895	
		3.376	0.557	0	-0.031	-286.96	-1.61E+01	0.00E+00	-2.3066249	-305.395	-2.00866	
		3.47	1.45	0	0.121	-294.95	-4.20E+01	0.00E+00	9.00327784	-327.932	-59.7469	
		5.457	1.045	0	-1.157	-463.845	-3.03E+01	0.00E+00	-86.0891939	-580.192	-72.5055	
		20.786	0.000123	-0.136	0	-1766.81	-3.56E-03	1.35E+00	0	-1765.47	-3.32635	
373.15												
		0	0	0	0	0	0.00E+00	0.00E+00	0	0	0	-329.5295741
403.15												
		3.28	0.593	0	0.04	344.4	2.18E+01	0.00E+00	-3.49420043	362.739	58.22472	
		3.639	0.506	0	-0.227	382.095	1.86E+01	0.00E+00	19.8295874	420.5546	10.17107	
		1.702	9.081	-2.164	0	178.71	3.34E+02	-2.81E+01	0	484.9098	0.971373	
		3.249	0.422	0	0.083	341.145	1.55E+01	0.00E+00	-7.25046589	349.4318	168.2664	
		3.376	0.557	0	-0.031	354.48	2.05E+01	0.00E+00	2.70800533	377.6958	2.484204	
		3.47	1.45	0	0.121	364.35	5.34E+01	0.00E+00	-10.5699563	407.1665	23.30009	
		0	0	0	0	0	0.00E+00	0.00E+00	0	0	0	
		20.786	0.000123	-0.136	0	2182.53	4.53E-03	-1.77E+00	0	2180.766	4.108828	267.5266902
											DH	**-1114.174889**
			OXIDATION									
	dh	-283										
403.15												
		3.28	0.593	0	0.04	-344.4	-2.18E+01	0.00E+00	3.49420043	-362.739	-58.2247	
		3.639	0.506	0	-0.227	-382.095	-1.86E+01	0.00E+00	-19.8295874	-420.555	-10.1711	
		1.702	9.081	-2.164	0	-178.71	-3.34E+02	2.81E+01	0	-484.91	-0.97137	
		3.249	0.422	0	0.083	-341.145	-1.55E+01	0.00E+00	7.25046589	-349.432	-168.266	
		3.376	0.557	0	-0.031	-354.48	-2.05E+01	0.00E+00	-2.70800533	-377.696	-2.4842	
		3.47	1.45	0	0.121	-364.35	-5.34E+01	0.00E+00	10.5699563	-407.167	-23.3001	
		20.786	0.000123	-0.136	0	-2182.53	-4.53E-03	1.77E+00	0	-2180.77	-4.10883	-267.5266902
423.15												
		3.28	0.593	0	0.04	410	2.67E+01	0.00E+00	-3.96315306	432.77	69.46569	
		3.639	0.506	0	-0.227	454.875	2.28E+01	0.00E+00	22.4908936	500.177	10.45184	
		1.702	9.081	-2.164	0	212.75	4.09E+02	-3.55E+01	0	586.5971	1.175073	
		3.249	0.422	0	0.083	406.125	1.90E+01	0.00E+00	-8.2235426	416.9257	200.7676	
		3.376	0.557	0	-0.031	422	2.51E+01	0.00E+00	3.07144362	450.1817	2.960963	
		3.47	1.45	0	0.121	433.75	6.54E+01	0.00E+00	-11.988538	487.1293	27.87596	
		20.786	0.000123	-0.136	0	2598.25	5.54E-03	-2.23E+00	0	2596.022	4.891222	317.5883444
											DH	**-61.87995385**

			pressure adsorption swing											
303.15														
		3.28	0.593	0	0.04		-16.4	-8.91E-01	0.00E+00	2.21E-01		-17.0701	-2.74	
		3.639	0.506	0	-0.227		-18.195	-7.61E-01	0.00E+00	-1.26E+00		-20.2114	-0.42234	
		1.702	9.081	-2.164	0		-8.51	-1.37E+01	9.78E-01	0.00E+00		-21.183	-0.04243	
		3.249	0.422	0	0.083		-16.245	-6.34E-01	0.00E+00	4.59E-01		-16.4202	-7.90704	
		3.376	0.557	0	-0.031		-16.88	-8.37E-01	0.00E+00	-1.71E-01		-17.8888	-0.11766	
		3.47	1.45	0	0.121		-17.35	-2.18E+00	0.00E+00	6.69E-01		-18.8603	-1.07928	
		20.786	0.000123	-0.136	0		-103.93	-1.85E-04	6.15E-02	0.00E+00		-103.869	-0.1957	-12.50445906
303.15														
		3.28	0.593	0	0.04		16.4	8.91E-01	0	-0.22127768		17.07015	2.739999	
		3.639	0.506	0	-0.227		18.195	7.61E-01	0	1.25575083		20.2114	0.422343	
		1.702	9.081	-2.164	0		8.51	1.37E+01	-0.978046913	0		21.18297	0.042434	
		3.249	0.422	0	0.083		16.245	6.34E-01	0	-0.45915119		16.42022	7.907039	
		3.47	1.45	0	0.121		17.35	2.18E+00	0	-0.66936498		18.86035	1.079283	
		20.786	0.000123	-0.136	0		103.93	1.85E-04	-0.061466904	0		103.8687	0.195701	
		3.376	0.557	0	-0.031		16.88	8.37E-01	0	0.1714902		17.8888	0.117659	12.50445906
			REACTOR										DH	**0**
	del Hr	-92												
303.15														
		3.28	0.593	0	0.04		-16.4	-2.97E-03	0	0.22127768		-16.1817	-14.3117	
		3.249	0.422	0	0.083		-16.245	-2.11E-03	0	0.45915119		-15.788	-41.8904	
		20.786	0.000123	-0.136	0		-103.93	-6.15E-07	0.061466904	0		-103.869	-1.07636	-57.27846514
723.15														
		3.28	0.593	0	0.04		16.4	2.97E-03	0	-0.22127768		16.18169	12.02183	
		3.249	0.422	0	0.083		16.245	2.11E-03	0	-0.45915119		15.78796	35.18795	
		29.75	0.00254	-1.226	755000		148.75	1.27E-05	-0.554106061	-41.7661621		106.4297	30.12176	
		20.786	0.000123	-0.136	0		103.93	6.15E-07	-0.061466904	0		103.8685	1.076355	78.4078901
													DH	**21.12942496**

PREHEATER	**(After De Sulpurisation)**															
	c	K														
Tin	300	573.15														
Tout	850	1123.15														
	Cp	A	B	C	D											
		1	1.00E-03	0.000001	100000											
CH4		1.702	9.081	-2.164	0		936.1	4.24E+03	-8.86E+02	0		4286.044	0.016063657	572.4151		
H2S		3.931	1.49	0	-0.232		2162.05	6.95E+02	0.00E+00	19.82187		2876.931	0.000496796	11.88277	**584.2979**	DH

PREHEATER	**(Before Secondry Reformer)**															
Tin	25	298.15														
Tout	500	773.15														
	Cp	A	B	C	D											
		1	1.00E-03	0.000001	100000											
O2		3.639	0.506	0	-0.227		1728.525	1.29E+02	0.00E+00	46.77576		1904.044	0.005197899	82.2839		
N2		3.28	0.593	0	0.04		1558	1.51E+02	0.00E+00	-8.24243		1700.637	0.019306481	272.9762		
INERT		20.786	0.123	-0.136	0		9873.35	3.13E+01	-1.97E+01	0		9884.896	0.00022662	18.62435	**373.8844**	DH

LER	**(Before Water gas shift reaction)**															
Tin	1000	1273.15														
Tout	350	623.15														
	Cp	A	B	C	D											
		1	1.00E-03	0.000001	100000											
N2		3.28	0.593	0	0.04		-2132	-3.65E+02	0.00E+00	3.277187		-2494.19	0.019306481	-400.352		
O2		3.639	0.506	0	-0.227		-2365.35	-3.12E+02	0.00E+00	-18.598		-2695.79	0.002908938	-65.1975		
CH4		1.702	9.081	-2.164	0		-1106.3	-5.60E+03	1.31E+03	0		-5388.86	0.000240943	-10.795		
H2		3.249	0.422	0	0.083		-2111.85	-2.60E+02	0.00E+00	6.800162		-2365.13	0.042888484	-843.345		
CO		3.376	0.557	0	-0.031		-2194.4	-3.43E+02	0.00E+00	-2.53982		-2540.22	0.015822135	-334.154		
H2O		3.47	1.45	0	0.121		-2255.5	-8.94E+02	0.00E+00	9.91349		-3139.22	0.036945023	-964.245		
INERT		20.786	0.123	-0.136	0		-13510.9	-7.58E+01	8.26E+01	0		-13504.1	0.00022662	-25.4434	**-2643.53**	DH

LER	**(Before CO2 removal)**												
Tin	350	623.15											
Tout	110	383.15											
	Cp	A	B	C	D								
		1	1.00E-03	0.000001	100000								
N2		3.28	0.593	0	0.04	-787.2	-7.16E+01	0.00E+00	4.020775	-854.788	0.019306481	-137.205	
O2		3.639	0.506	0	-0.227	-873.36	-6.11E+01	0.00E+00	-22.8179	-957.28	0.002908938	-23.1517	
CH4		1.702	9.081	-2.164	0	-408.48	-1.10E+03	1.34E+02	0	-1371.09	0.000240943	-2.74657	
H2		3.249	0.422	0	0.083	-779.76	-5.10E+01	0.00E+00	8.343109	-822.376	0.057919512	-396.009	
CO		3.376	0.557	0	-0.031	-810.24	-6.73E+01	0.00E+00	-3.1161	-880.617	0.000791107	-5.79205	
H2O		3.47	1.45	0	0.121	-832.8	-1.75E+02	0.00E+00	12.16285	-995.733	0.021913995	-181.416	
CO2		5.457	1.045	0	-1.157	-1309.68	-1.26E+02	0.00E+00	-116.301	-1552.17	0.015031029	-193.972	
INERT		20.786	0.123	-0.136	0	-4988.64	-1.49E+01	8.42E+00	0	-4995.07	0.00022662	-9.41133	**-949.704**
LER	**(Before Pressure swing adsorption)**												
Tin	150	423.15											
Tout	30	303.15											
N2		3.28	0.593	0	0.04	-393.6	-2.58E+01	0.00E+00	3.741875	-415.7	0.019306481	-66.7257	
O2		3.639	0.506	0	-0.227	-436.68	-2.21E+01	0.00E+00	-21.2351	-479.966	0.002513384	-10.0295	
CH4		1.702	9.081	-2.164	0	-204.24	-3.96E+02	3.46E+01	0	-565.414	0.000240943	-1.13264	
H2		3.249	0.422	0	0.083	-389.88	-1.84E+01	0.00E+00	7.764391	-400.506	0.057919512	-192.861	
CO2R		3.376	0.557	0	-0.031	-405.12	-2.43E+01	0.00E+00	-2.89995	-432.293	0.000791107	-2.8433	
H2O	-294.590939	3.47	1.45	0	0.121	-416.4	-6.32E+01	0.00E+00	11.31917	-468.269	0.006882966	-26.7967	
INERT		20.786	0.123	-0.136	0	-2494.32	-5.36E+00	2.17E+00	0	-2497.51	0.00022662	-4.70561	**-599.685**
LER	**(After Reactor)**												
Tin	450	723.15											
Tout	-20	253.15											
N2		3.28	0.593	0	0.04	-1541.6	-1.36E+02	0	10.26955	-1667.38	0.106379357	-1474.7	
H2		3.249	0.422	0	0.083	-1527.03	-9.68E+01	0	21.30932	-1602.54	0.319138072	-4252.04	
NH3	-799.972767	29.75	2.54	-1.226	7.55	-13982.5	-5.83E+02	147.915	1938.378	-13278.9	0.034041394	-3758.21	
INERT		20.786	0.123	-0.136	0	-9769.42	-2.82E+01	16.40818	0	-9781.23	0.001246412	-101.36	**-10386.3**
total energy change		-11584											

8. REFERENCES:

[1] Erdemir, Dogan, Dincer, Ibrahim, *A perspective on the use of ammonia as a clean fuel: challenges and solutions,* International Journal of energy Research, 2021.

[2] Ghavam, Seyedehhoma, *Sustainable ammonia production processes,* Frontiers in Energy Research, 2021.

[3] Kissel, David E, *The historical development and significance of the Haber Bosch Process, 2014.*

[4] Liu, Huazhang, *Ammonia synthesis catalyst 100 years: Practice, enlightenment and challenge, 2014.*

[5] Qi, Gongshin, *Superior sorbent for natural gas desulfurization,* Industrial & engineering chemistry research, 2008.

[6] Valeh-e-Sheyda, Peyvand, *Integration of commercial CO2 capture plant with primary reformer stack of ammonia plant.* Journal of Thermal Analysis and Calorimetry, 2019.

[7] Yu, Y-H, *Simulation of secondary reformer in industrial ammonia plant* ,Chemical engineering & technology, 2002.

[8] RJ, Byron Smith, *International journal of Chemical Reactor Engineering,* 2010.

[9] Gao, Hongwei, *First Principles study on the adsorption of CO_2 and H_2O on the K_2CO_3 surface,* Surface Science, 2013.

[10] Royer, Sebastien, *Catalytic oxidation of carbo monoxide over transition metal oxides, 2011.*

[11] Sircar, Shivaji, *pressure swing adsorption,* Industrial & engineering chemistry research, 2002.

[12] Robso, Jaochin Weel, *Optimisation and Robust Control of a Load-Flexible Haber-Bosch Ammonia Synthesis Loop,* 2023.

[13] Modak, Jayant M, *Haber process for ammonia synthesis, 2002.*

[14] J.M. Smith, *Introduction to Chemical Engineering Thermodynamics,* Mc Grill Education,2017.